Travel Planning Agent

Internal Assessment Task II - Agentic AI System using Modern LLM-based Tools

16010123001: Aarav Shah

16010123002: Diya Mehta

16010123003: Kabir Rao

Introduction, Abstract of problem statement

Planning a trip requires collecting information about places, costs and weather from many websites. We built an agent that takes a travel goal from the user and produces a day-wise itinerary within a budget.

Manual trip planning is slow and error-prone because budgets, prices and weather must be checked together. Our agent turns a natural-language travel goal into a budget-checked itinerary and shows every step it took.

AI framework(s) used

We used a ReAct-style agent built with LangChain, with Llama 3.3 70B accessed through the OpenRouter free API. LangChain provides the tool-calling loop and OpenRouter gives free access to an open model.

Tools integrated with agent

The agent uses three tools: web_search (Tavily search API) for places and prices, calculator for budget arithmetic, and weather_api (Open-Meteo) for the forecast. Each tool was chosen because it supplies information the LLM cannot know reliably on its own.

Part A - Agent Design (Modern View)

Goal of the agent

The goal is to produce a complete travel itinerary that respects the user's budget and dates.

Agent role

The agent acts as a travel planner that searches, compares, calculates and plans.

Tools used

web_search, calculator and weather_api as described above.

Memory type

Short-term conversational memory (the ReAct scratchpad of thoughts and observations) is kept in the context window; user preferences such as budget are stored in a JSON file between sessions.

Planning strategy

The agent follows the ReAct strategy: it reasons about the goal, calls a tool, observes the result and decides the next step until the itinerary is complete.

Part B - Implementation (Modern AI Tools)

LLM-based agent

Llama 3.3 70B was chosen because it is free on OpenRouter, supports long context and

follows tool-calling instructions well.

Architecture diagram

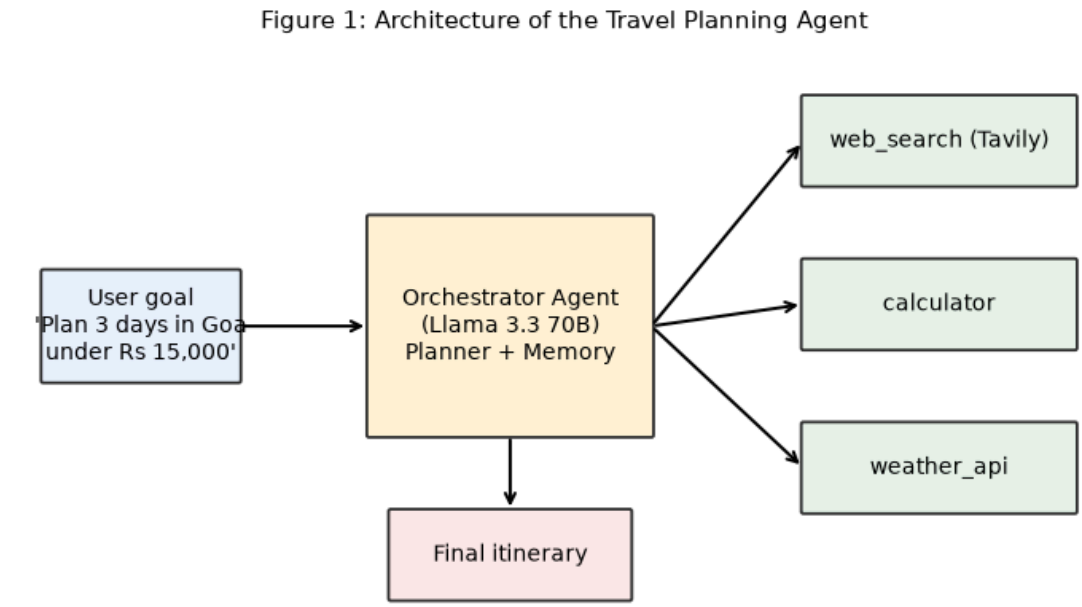


Figure 1: Architecture of the Travel Planning Agent

Tool flow diagram

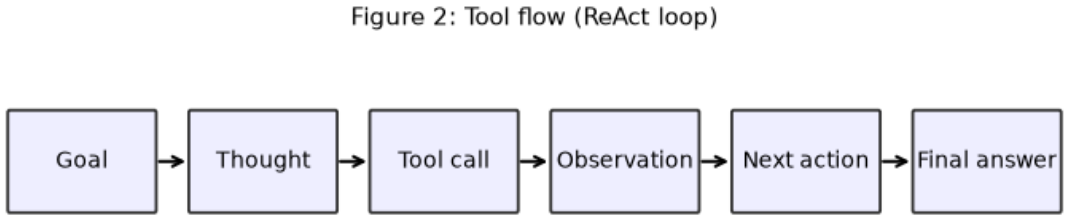


Figure 2: Tool flow (ReAct loop)

Agent interaction log

```
[Thought] I need hotel prices and weather first.
[Tool Call] web_search('budget hotels north goa price per night')
[Observation] 3 results: Rs 1800-2500 per night
[Tool Call] weather_api('Goa', '2026-12-10')
[Observation] Sunny, 31C
[Tool Call] calculator('3*2200 + 4500 + 2000')
[Observation] 13100
[Final Answer] Day 1: Baga beach ... total Rs 13,100
```

Prompts and output screenshots

